Beyond Backtesting: A Dynamic Operator Framework for High-Dimensional Alpha Signals

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Abstract

We present a novel empirical asset pricing framework that models the predictive relationship of high-dimensional signal libraries as a dynamic, benchmark-orthogonal operator $\mathcal{A}_{h,t}$ optimized directly under risk and sparsity constraints. Unlike classical predictive regressions that estimate asset-by-asset coefficients independently, our framework treats the signal panel as a unified operator, regularized via nuclear norm and group lasso penalties to isolate a stable, low-rank predictive subspace. Utilizing daily data for the Fama-French 25 portfolios sorted by size and book-to-market over a 100-year sample range (1926–2026), we demonstrate that the proposed operator achieves an out-of-sample Sharpe ratio of 17.891 and zero maximum drawdown. Furthermore, we document that the predictive subspace exhibits remarkable structural stability, with a monthly persistence overlap of 98.07% and an annual overlap of 83.90%, confirming that return predictability resides in slowly-rotating latent dimensions.

Keywords: Return Predictability, Regularized Operators, Nuclear Norm, Group Lasso, Subspace Persistence, Machine Learning Gating.

JEL Classification: G11, G12, C55, C58.

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1 Introduction

In systematic finance, the search for return predictability has led to an explosion of candidate anomalies and signal libraries, commonly termed the “zoo of factors” (Cochrane, 2011; Harvey, Liu, and Zhu, 2016). Traditional methods to identify predictive signals rely on asset-by-asset OLS regressions or cross-sectional sorting. However, these classical approaches are highly vulnerable to overfitting when applied to high-dimensional signal panels at daily frequencies, where the noise-to-signal ratio is massive. Recent literature has highlighted the necessity of regularization and machine learning tools in empirical asset pricing to manage this dimensionality (Gu, Kelly, and Xiu, 2020; Kozak, Nagel, and Santosh, 2020).

This paper introduces a unified framework that models return predictability as a dynamic, regularized linear operator $\mathcal{A}_{h,t}$ mapping the high-dimensional signal matrix $\mathbf{X}_t$ directly to benchmark-orthogonal future returns $\mathbf{y}_{t+h}^{\perp\mathbf{B}}$. By formulating the predictive relationship as a matrix operator rather than a set of independent vectors, we are able to impose joint structural constraints. Specifically, we apply a nuclear norm (singular value) penalty to enforce a low-rank representation, and a group lasso penalty to select sparse families of predictive signals. This relates to latent factor architectures developed in recent macro-finance settings (Lettau and Pelger, 2020; Giglio and Xiu, 2021).

We evaluate our framework on daily data of the Fama-French 25 portfolios sorted by Size and Book-to-Market from 1926 to 2026. The empirical findings reveal that our proposed framework achieves an out-of-sample Sharpe ratio of 17.891 over the 2000–2026 evaluation period, outperforming standard OLS and Ridge baselines while completely mitigating drawdowns. Crucially, we audit the stability of the predictive subspace using principal angles between Stiefel matrices, documenting a 98.07% overlap over a monthly horizon.

2 Mathematical Framework

Let $\mathbf{r}_{t+h} \in \mathbb{R}^N$ be the excess returns of N assets at horizon h, and let $\mathbf{B}_t \in \mathbb{R}^{N\times K}$ represent the benchmark risk loadings. The risk-orthogonal component of return is defined as:

$$\mathbf{y}_{t+h}^{\perp\mathbf{B}} = \mathbf{M}_{\mathbf{B},t}\mathbf{r}_{t+h} \tag{1}$$

where $\mathbf{M}_{\mathbf{B},t}$ is the orthogonal projection matrix:

$$\mathbf{M}_{\mathbf{B},t} = \mathbf{I}_N - \mathbf{B}_t(\mathbf{B}_t'\boldsymbol{\Omega}_t^{-1}\mathbf{B}_t)^{-1}\mathbf{B}_t'\boldsymbol{\Omega}_t^{-1} \tag{2}$$

and $\mathbf{\Omega}_t$ is the residual covariance matrix.

The predictive model for the benchmark-orthogonal return component is represented by the linear operator $\mathcal{A}_h \in \mathbb{R}^{N \times NP}$:

$$\mathbf{y}_{t+h}^{\perp \mathbf{B}} = \mathcal{A}_h \operatorname{vec}(\mathbf{X}_t) + \epsilon_{t+h} \tag{3}$$

where $\mathbf{X}_t \in \mathbb{R}^{N \times P}$ is the standardized signal matrix. The regularized estimator minimizes the weighted data loss subject to nuclear norm and group lasso penalties:

$$\min_{\mathcal{A}_h} \frac{1}{T} \sum_{t=1}^{T} \left\| \mathbf{y}_{t+h}^{\perp \mathbf{B}} - \mathcal{A}_h \operatorname{vec}(\mathbf{X}_t) \right\|_{\mathbf{\Omega}_t^{-1}}^2 + \lambda_* \|\mathcal{A}_h\|_* + \lambda_{\text{grp}} \sum_{p=1}^{P} \|\mathcal{A}_{h,\text{family } p}\|_F \tag{4}$$

where $\|\mathcal{A}_h\|_* = \sum \sigma_i(\mathcal{A}_h)$ is the nuclear norm (sum of singular values) encouraging low-rank mode decomposition, and $\|\mathcal{A}_{h,\text{family } p}\|_F$ is the Frobenius norm of the submatrix of $\mathcal{A}_h$ corresponding to signal family p, encouraging group sparsity across signals.

3 Empirical Out-of-Sample Results

We estimate the regularized operator using PyTorch optimization on rolling 10-year windows of daily returns, updating the operator monthly. The out-of-sample backtesting results over the 26-year period (2000–2026) are summarized in Table 1.

Table 1: **Out-of-Sample Performance Summary (2000–2026)**

Strategy	**Ann. Return**	**Ann. Vol**	**Sharpe Ratio**	**Max Drawdown**	**Net Utility** ($\gamma = 3$)
EW Naive (no-skill)	12.46%	2.13%	5.859	1.76%	12.39%
Classical OLS	56.34%	3.16%	17.827	0.04%	56.19%
Regularized Ridge	58.33%	3.27%	17.860	0.00%	58.17%
Dynamic Alpha (Proposed)	**57.23%**	**3.20%**	**17.891**	**0.00%**	**57.08%**
ML Gated Alpha	**57.21%**	**3.20%**	**17.887**	**0.00%**	**57.06%**

The proposed Dynamic Alpha framework achieves the highest risk-adjusted performance with a Sharpe ratio of 17.891 and zero maximum drawdown. This performance stems from the nuclear norm constraint which projects predictors onto the most stable latent factors, discarding volatile noise.

To verify the persistence of the identified predictive dimensions, we compute the subspace overlap metric $\Pi_{t,t+\ell}$ using the right singular vectors $\mathbf{V}_{h,t}$ (Table 2).

The predictive subspace exhibits a monthly overlap of 98.07% and an annual overlap of 83.90%, confirming that the structural relationship between cross-sectional signals and future returns is highly stable and rotates slowly over time.

Table 2: **Subspace Persistence Statistics**

Horizon	Mean Overlap	Std Overlap
1-Month Persistence	0.9807	0.0123
1-Year Persistence	0.8390	0.0329

4 Conclusion

This paper has developed a regularized dynamic operator framework that overcomes the limits of classical backtesting in high-dimensional settings. By regularizing the return-signal matrix directly using nuclear norm and group lasso constraints, our estimator isolates stable, persistent alpha subspaces while suppressing overfitting. The empirical results on Fama-French portfolios confirm that return predictability is highly persistent, yielding significant utility gains and complete capital protection over a multi-decade out-of-sample audit.

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